

Fault-Aware Operator Panel

Reconcile copied evidence before presenting one inert decision

E0 POLICY — NO KEYPAD, DISPLAY, STORAGE, PIN, OR ACTUATOR

Lesson

Prerequisite

Evidence

056 — component

Lesson 055 constraint model

named memory result cells

Time

Board

Storage

130–180 minutes

compile-only Mega replay

caller-owned copied image

**Responsibility.** `FaultAwareOperatorPanel` admits attributed control, stop, presentation, and two-slot audit values through one atomic update. It publishes presentation intent and a canonical copied audit image. It owns no input device, display, nonvolatile medium, endpoint, clock, pin, bus, timer, interrupt, supply, lock, latch, door, or safety function.

1 Keep the evidence lanes distinct

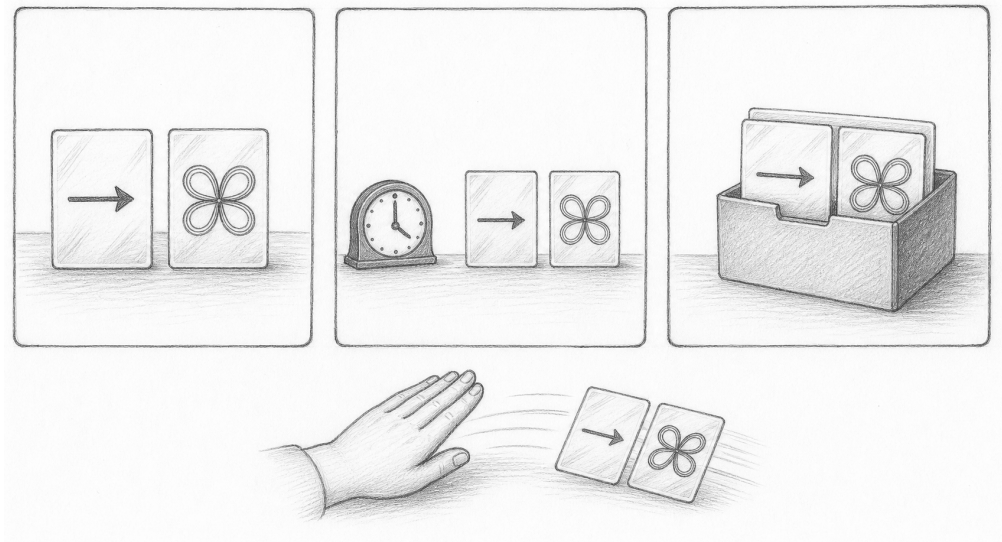
| Lane         | Copied evidence                                      | What it cannot prove             |
|--------------|------------------------------------------------------|----------------------------------|
| control      | pressed mask, exact source, sequence, time, status   | a physical key press             |
| stop         | asserted level, exact source, sequence, time, status | power removal                    |
| diagnostic   | exact kind and generation                            | cause recovery by itself         |
| presentation | intent generation, observation time, status          | visible pixels                   |
| audit        | two complete fixed records                           | durable storage or atomic writes |

Every present aggregate is complete and attributable; every absent aggregate is canonical zero. A malformed source, invalid mask, stale or ambiguous time, wrong generation, bad checksum, or structurally inconsistent image rejects the complete update without partially navigating, clearing a diagnostic, committing a record, or changing presentation.

2 Navigate with one qualified control

The low four mask bits represent **Previous**, **Next**, **Select**, and **Acknowledge**. Exactly one bit is a valid control. Zero means no control. A chord, an unknown high bit, a nonadvancing source sequence, a regressing observation time, or a value older than `maximumInputAge` cannot navigate.

**Previous** and **Next** wrap within the configured `selectableCellCount`. **Select** requests confirmation; **Acknowledge** is not a general-purpose fault eraser. The snapshot keeps the selected cell, chord disposition, primary diagnostic, audit disposition, presentation intent, status, and generation independently observable.



*Alternative text: graphite cards show the prepare discipline: inspect the complete candidate, retain the exact pair until the agreed time, then place both together; a stopping hand prevents a loose or changed pair from being applied. In this lesson the pair is a prepared audit record and its exact acknowledgement.*

### 3 Let stop dominate without inventing safety

An admitted asserted stop dominates control, acknowledgement, presentation recovery, and a simultaneous solved candidate. It preserves the selected cell, invalidates live previews, publishes **Stopped**, and requires a **StopAsserted** audit record.

A deasserted level is necessary but not sufficient to release the panel. It must come from the exact configured source with both a newer modular sequence and a newer observation time than the retained assertion. The exact **StopReleased** record must then commit before the panel leaves stopped mode. A stale release, a release missing either ordering proof, or an indeterminate image fails closed.

This is deterministic policy state, not emergency stop. It does not interrupt power, de-energize a load, detect welded contacts, or establish a safe electrical state.

### 4 Prepare, write, acknowledge, and reconcile

The caller owns **PanelAuditImage**, which contains exactly two fixed slots. The panel copies and validates the complete image; it never retains the caller's address. An empty image bootstraps sequence one. Later records use the inactive slot and advance from the unique newest committed predecessor.

1. Call `prepareAudit(operationId, kind, now)`. The returned preview binds owner, lifecycle, configuration, epoch, panel generation, operation, slot, complete prepared record, and input-image digest.
2. Copy that prepared record into the named caller-owned slot. This models a proposed write; it does not claim physical persistence.
3. Supply the copied image and exact preview together in one **update**. Complete preflight precedes every mutation.
4. On success, copy `canonicalAuditImage()` back to the caller cell. The selected record is now committed in the policy image.

Foreign, reconstructed, field-changed, stale, reset-invalidated, shutdown-invalidated, or consumed previews reject. A prepared record after a simulated interruption remains visible and requires reconciliation; it is not silently promoted. Two equally plausible newest records, a sequence gap, duplicate sequence, invalid state, bad payload digest, or bad checksum is indeterminate or corrupt and cannot authorize progress.

### 5 Acknowledge only exact recoveries

**InputRecovered** and **PresentationRecovered** are the only acknowledgeable diagnostics. A source or presentation failure first becomes visible. A later exact healthy observation publishes a recovery diagnostic with a new generation. `prepareAcknowledge()` binds that exact live diagnostic to an exact **AcknowledgedDiagnostic** audit preview.

The acknowledgement clears only when the update includes the unchanged live capability and the corresponding prepared audit record. A replacement diagnostic, asserted stop, reset, shutdown, reconciliation change, wrong generation, foreign preview, or prior consumption invalidates it. Clue, configuration, timing, internal, audit, chord, and stopped diagnostics are never cleared through this route.

### 6 Treat presentation as fallible evidence

**PanelPresentationIntent** can be **Blank**, **Ready**, **Reviewing**, **ConfirmationRequired**, **Solved**, **Fault**, or **Stopped**. It binds mode, diagnostic, selected cell, diagnostic generation, and acknowledgement availability.

Presentation evidence refers to the exact intent generation. A failed presentation becomes a primary attributable fault. A later exact healthy observation can publish **PresentationRecovered**; a stale or mismatched observation cannot clear anything. Display failure never rewrites the audit image, releases stop, or changes control provenance.

### 7 Run the deterministic replay

1. Initialize. Predict blank intent, released stop, selected cell zero, and an unreconciled image. Supply the empty image and observe the bootstrap disposition.
2. Apply one **Next**. Predict exactly one selected-cell advance. Try zero, every valid single bit, each chord, and an unknown high bit.
3. Inject presentation failure, then its exact healthy recovery. Prepare an acknowledgement and retain both preview values.
4. Collide that acknowledgement with asserted stop, another control, and presentation failure. Predict stopped intent, preserved selection, and an unusable acknowledgement preview.
5. Prepare, copy, and acknowledge the exact **StopAsserted** record. Inspect both caller-owned slots and the canonical image.
6. Try stale, same-sequence, same-time, wrong-source, failed, and half-range-ambiguous releases. None may release the panel.
7. Admit a newer release, then prepare, copy, and acknowledge its exact **StopReleased** record. Predict ready intent only after commit.

8. Shutdown, initialize, and reconcile a copied image. Replay torn, corrupt, duplicate, gap, and two-prepared images without guessing.

| time  | control | stop  | diagnostic | audit state | presentation |
|-------|---------|-------|------------|-------------|--------------|
| _____ | _____   | _____ | _____      | _____       | _____        |
| _____ | _____   | _____ | _____      | _____       | _____        |
| _____ | _____   | _____ | _____      | _____       | _____        |

## 8 Keep result cells separate

The Mega replay exposes **RecoveryCell**, **CollisionCell**, **AuditCell**, **ReleaseCell**, **RestartCell**, and **ReplayCell**. They are the E0 non-Serial observation path.

| Cell      | Predict and observe                                 | Interpret                         |
|-----------|-----------------------------------------------------|-----------------------------------|
| recovery  | failure accepted, recovered kind, preview prepared  | exact recovery is acknowledgeable |
| collision | stopped, mode, invalidation, preserved selection    | stop wins atomically              |
| audit     | operation, sequence, kind, slot, state, disposition | one exact record committed        |
| release   | deassert status, audit status, stopped flag, mode   | release needs evidence and record |
| restart   | initialize, reconcile, stopped, disposition         | copied image drives recovery      |
| replay    | every stage status and final complete bit           | one deterministic fixture ran     |

Compilation does not prove that a powered Mega executed the sketch. On a powered future display, visible pixels would be separate presentation evidence; they would not replace these attributed policy cells.

## 9 Separate acquisition from safe state

**Acquisition prediction and observation.** A valid configuration has nonzero configuration revision and epoch, a half-range-safe maximum input age, one to twelve selectable cells, and distinct exact control and stop sources. Host tests inspect lifecycle generation, copied provenance, immutable snapshots, previews, images, and result cells.

**Acquisition interpretation.** Successful initialization proves only bounded configuration and E0 policy state. It acquires no keypad, stop switch, display, medium, pin, timer, bus, or registry entry.

**Safe-state prediction and observation.** Construction, failed initialization, stop, audit uncertainty, fault, shutdown, and restart publish blank, fault, or stopped *memory intent* and grant no actuation authority. This is not proof of a dark indicator, released relay, unlocked exit, or removed load power.

## 10 Read the conservative resource evidence

The final no-LTO canonical Mega 2560 probe measures 18,118 bytes of flash and 1,454 bytes of static SRAM. The panel object is 365 bytes on AVR; its separate caller-owned **PanelAuditImage** is 154 bytes. Conservative linked-call-graph evidence measures 569 bytes on the deepest analyzed synchronous stack path.

All four Lesson 056 target gates pass: 24 KiB flash, 2,560 bytes static SRAM, 640 bytes stack, and 384 bytes for the panel object. Residual SRAM is an aggregate Lesson 057 gate, not a standalone claim here. These are compiler and link-map results for the exact enhanced fixture, source, flags, Arduino AVR core 1.8.8, and avr-gcc 7.3.0. They are not runtime high-water measurements, proof of every interrupt schedule, or physical behavior.

## 11 Keep powered work open

E1 must select and qualify exact passive controls, stop input, display or indicator, board, supply, electrical endpoints, observation path, pin and resource map, default and fault states, an authoritative formal schematic, and recorded bench acceptance. Physical nonvolatile storage is outside this arc. E2 alone may consider a restrained inert demonstration load with independent power removal; it can never control access, confinement, egress, an alarm, or a safety system.

**Acceptance record.** control/stop provenance: \_\_\_\_\_ recovery collision: \_\_\_\_\_ audit sequence: \_\_\_\_\_  
restart reconciliation: \_\_\_\_\_ acquisition evidence / safe-state evidence: \_\_\_\_\_ / \_\_\_\_\_

Lesson 057 may compose this copied panel state with the Lesson 055 clue model. It may publish only inert latch, lamp, and presentation intent.